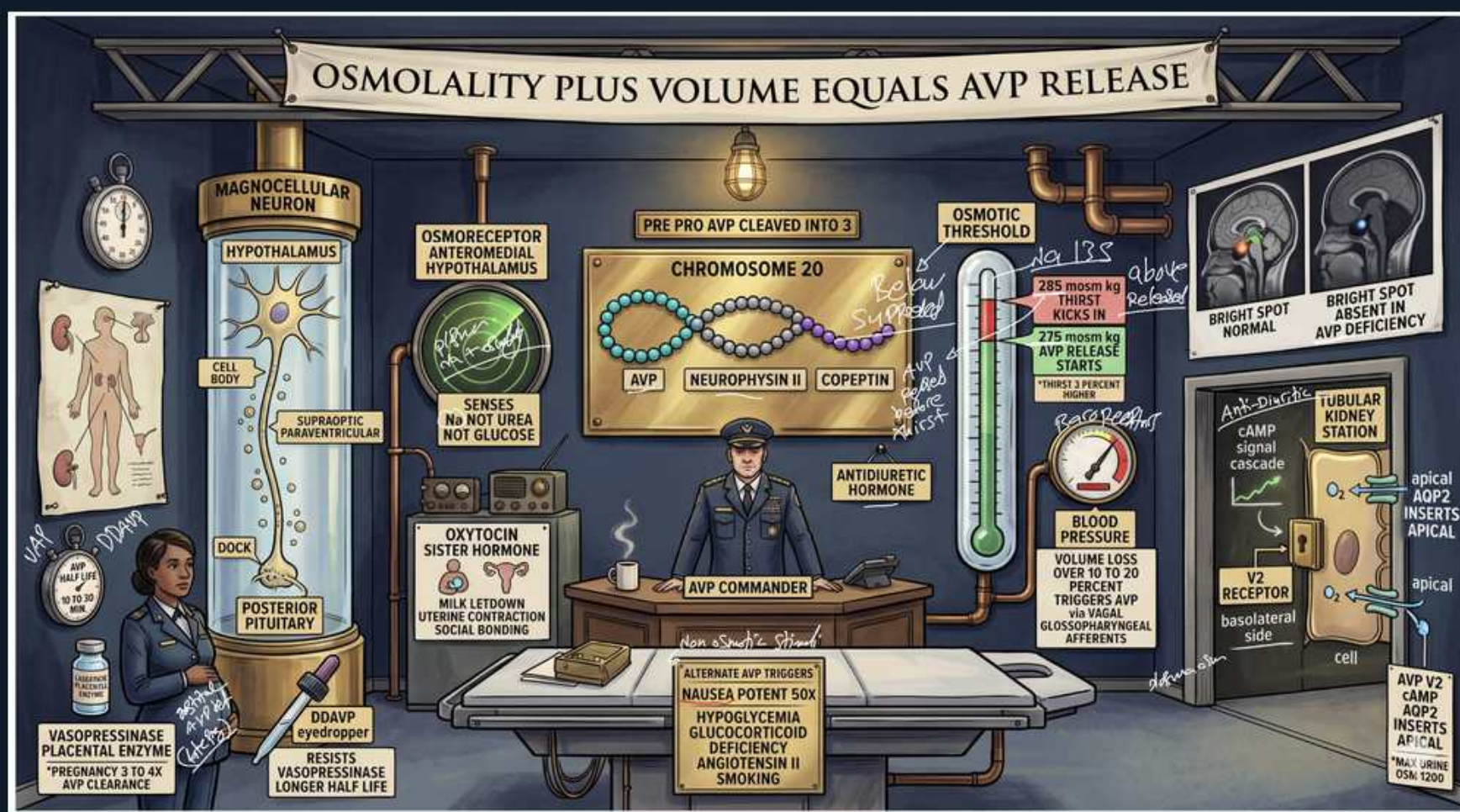


AVP (Vasopressin) Physiology — Synthesis to V2 Receptor



1. Osmolality + volume → AVP release

WHAT YOU SEE

Banner 'OSMOLALITY PLUS VOLUME EQUALS AVP RELEASE'.

WHAT IT ENCODES

AVP secretion is driven by two inputs: plasma osmolality (sensed by hypothalamic osmoreceptors, exquisitely sensitive) and effective blood volume/pressure (baroreceptors, less sensitive but dominant when severe). Both converge on the supraoptic/paraventricular neurons.

BOARD TRAP

Osmotic control is the fine-tuner; a large volume/BP drop overrides it and releases AVP even when the patient is hypo-osmolar — the basis of hypovolemic hyponatremia.

2. Magnocellular neuron / hypothalamus

WHAT YOU SEE

Large neuron, 'MAGNOCELLULAR NEURON', hypothalamus, cell body.

WHAT IT ENCODES

AVP is synthesized in magnocellular neurons of the supraoptic and paraventricular nuclei of the hypothalamus, then transported down axons. Damage here (not the gland) causes central DI.

BOARD TRAP

Central DI is a hypothalamic/stalk problem — a lesion sparing the magnocellular neurons but hitting only the gland tip may cause only transient DI.

3. Posterior pituitary storage

WHAT YOU SEE

'POSTERIOR PITUITARY' storage dock.

WHAT IT ENCODES

AVP travels along the supraopticohypophyseal tract and is STORED/released from the posterior pituitary (neurohypophysis). The stored hormone produces the T1 MRI 'bright spot'.

BOARD TRAP

The posterior pituitary stores but does not synthesize AVP — high stalk-section lesions abolish the bright spot; low ones may spare it.

4. Osmoreceptor — senses Na, not glucose/urea

WHAT YOU SEE

'OSMORECEPTOR anteromedial hypothalamus — senses Na not urea, not glucose'.

WHAT IT ENCODES

Hypothalamic osmoreceptors respond to EFFECTIVE osmoles (sodium). Urea and glucose (in the absence of insulin deficiency) cross cell membranes and are ineffective osmoles that don't trigger AVP.

BOARD TRAP

A high BUN raises measured osmolality but is an INEFFECTIVE osmole — it doesn't stimulate AVP or shift water, so don't treat urea-driven hyperosmolality like hyponatremia.

5. Chromosome 20 — AVP-neurophysin II-copeptin

WHAT YOU SEE

DNA 'CHROMOSOME 20', pre-pro cleaved into 3: AVP, neurophysin II, copeptin.

WHAT IT ENCODES

The AVP gene on chromosome 20 makes a preprohormone cleaved into AVP, neurophysin II (carrier), and copeptin. Copeptin's 1:1 co-secretion underlies its use as an AVP surrogate.

BOARD TRAP

Copeptin is co-released stoichiometrically with AVP — that is exactly why a copeptin assay can stand in for the unmeasurable AVP.

6. Osmotic threshold ~280–285

WHAT YOU SEE

Thermometer 'OSMOTIC THRESHOLD' — thirst kicks in, AVP starts.

WHAT IT ENCODES

AVP secretion begins around plasma osmolality 280–285 mOsm/kg; thirst is triggered slightly higher (~290–295). This narrow range keeps osmolality tightly regulated.

BOARD TRAP

Thirst threshold sits ABOVE the AVP threshold — losing thirst (e.g., adipsic/hypothalamic lesions) is more dangerous because the last defense against hyponatremia is gone.

7. Blood pressure / volume — 5–10% drop

WHAT YOU SEE

Gauge 'BLOOD PRESSURE — volume loss over 5 to 20 percent... via vagal afferents'.

WHAT IT ENCODES

Baroreceptor-mediated AVP release requires a larger stimulus (roughly >5–10% volume/BP fall) but then becomes powerful, overriding osmotic suppression. Signals via vagal/glossopharyngeal afferents.

BOARD TRAP

Small volume changes don't move AVP, but large hypovolemia massively stimulates it — explaining ADH-mediated water retention and hyponatremia in heart failure/cirrhosis.

8. V2 receptor → cAMP → AQP2

WHAT YOU SEE

Right inset: V2 receptor, basolateral, cAMP, AQP2 inserts apical.

WHAT IT ENCODES

AVP binds the basolateral V2 receptor on collecting-duct principal cells, activating Gs/cAMP/PKA, which inserts aquaporin-2 (AQP2) channels into the apical membrane to reabsorb water. Nephrogenic DI is a defect here.

BOARD TRAP

Nephrogenic DI = V2/AQP2 defect downstream of normal AVP — so DDAVP (a V2 agonist) fails, distinguishing it from central DI.

9. Bright spot present vs absent

WHAT YOU SEE

Two MRI insets: bright spot normal vs absent in AVP deficiency.

WHAT IT ENCODES

The posterior pituitary T1 bright spot represents stored AVP-neurophysin; it is present normally and absent in central (AVP-deficient) DI. Used to localize central causes on MRI.

BOARD TRAP

Absence of the bright spot supports central DI but can also fade with normal aging — interpret alongside copeptin, not in isolation.

10. Oxytocin — sister hormone

WHAT YOU SEE

'OXYTOCIN SISTER HORMONE — uterine contraction, milk letdown, bonding'.

WHAT IT ENCODES

Oxytocin is co-produced in the same magnocellular system; it drives uterine contraction, milk letdown, and social bonding. Structurally similar to AVP, explaining minor cross-reactivity.

BOARD TRAP

High-dose oxytocin has weak antidiuretic (V2) activity and can cause water intoxication/hyponatremia during labor inductions.

11. V1 effects — vasoconstriction / nausea / hypoglycemia

WHAT YOU SEE

AVP commander desk; 'alternate AVP triggers — nausea, hypoglycemia, angiotensin II, smoking'.

WHAT IT ENCODES

Beyond osmolality/volume, AVP release is stimulated by nausea (most potent), pain, hypoglycemia, angiotensin II, and nicotine. V1 receptors mediate vasoconstriction and glycogenolysis.

BOARD TRAP

Nausea is one of the strongest non-osmotic AVP stimuli — post-op nausea/pain drives SIADH-like hyponatremia even with normal osmoreceptors.

12. Desmopressin (DDAVP) — V2-selective, long half-life

WHAT YOU SEE

Bottom-left 'DDAVP — V2 selective, resists vasopressinase, longer half-life'.

WHAT IT ENCODES

Desmopressin is a synthetic V2-selective analog with minimal V1 pressor effect, resistance to vasopressinase, and a longer half-life — making it the treatment for central DI and a tool in the DDAVP concentration test.

BOARD TRAP

DDAVP's V2 selectivity means it won't help nephrogenic DI (receptor/channel resistance) — and overuse risks dilutional hyponatremia.

13. Vasopressinase — pregnancy / placenta

WHAT YOU SEE

Bottom-left 'VASOPRESSINASE placental enzyme, increases AVP clearance 3–4x'.

WHAT IT ENCODES

Placental vasopressinase degrades native AVP, increasing clearance 3–4x and causing transient gestational DI in the 3rd trimester. DDAVP (vasopressinase-resistant) is the treatment.

BOARD TRAP

Gestational DI is treated with DDAVP, not native vasopressin, because placental vasopressinase rapidly destroys endogenous/native AVP.
